

Nikita Thakur

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EDUCATION

University of Illinois at Chicago (UIC) - M.S. in Computer Science

May 2023

Mumbai University - B.E. in Computer Engineering

May 2017

TECHNICAL SKILLS

Leadership: Team Management, Project Coordination, Vendor Management, SLA Enforcement, Stakeholder Engagement

Healthcare Systems: EPIC Certification, Healthcare Integrations, HL7, NCPDP, EDI Standards

Languages: Python, Java, C++, R, TypeScript, JavaScript

ML/AI: TensorFlow, PyTorch, Scikit-Learn, NLTK, Transformers, NLP, Deep Learning, AI Agents, Workflow Automation

Data: Pandas, NumPy, PySpark, Azure Data Factory, Matplotlib, Seaborn, D3.js

Cloud & DevOps: AWS (Lambda, API Gateway, DynamoDB), Azure ML, Docker, Kubernetes

Databases: MongoDB, PostgreSQL, MS SQL, Oracle 12g

Regulatory: Experience with compliance frameworks and regulatory requirements

PROFESSIONAL EXPERIENCE

Research Data Analyst, University of Illinois Cancer Center

Chicago, IL | January 2024 – Present

- Led NLP-driven healthcare project with intelligent AI agents extracting Social Determinants of Health from 50,000+ clinical notes.
- Engineered automated AI workflows and ETL pipelines with Azure Data Factory reducing data integration time by 65% across relational and document databases
- Developed ML-enhanced Python application for Breast Cancer Screening team with automated risk stratification, reducing follow-up delays and saving 15+ hours weekly
- Coordinated with clinical stakeholders to gather requirements and implemented healthcare data integrations with adherence to regulatory standards
- Collaborated with Business Intelligence team to design and implement reporting solutions for key healthcare metrics.

Full Stack Software Engineering Intern, Slalom Build

Chicago, IL | June 2022 – August 2022

- Architected **serverless** event management platform using React, TypeScript, Node.js and **AWS services**
- Built high-performance **REST APIs** with Lambda and API Gateway, improving request efficiency.
- Implemented responsive UI components with Material UI and state management with **Redux**
- Optimized data fetching with Axios interceptors, reducing unnecessary API calls.

Senior Software Engineer, Accenture

Mumbai, India | October 2017 – July 2021

- **Managed an 8-person Monitoring team** responsible for enterprise integration systems, providing leadership in task allocation, performance management, and professional development
- Led technical integration efforts between enterprise systems using IBM Sterling Integrator, handling secure file transfers via FTP, SFTP, AS2, OAuth, and HTTP protocols
- Designed and implemented system monitoring solutions with 99.9% uptime, ensuring compliance with Service Level Agreements (SLAs)
- Maintained vendor relationships and enforced SLAs, negotiating technical requirements and resolving escalated issues
- Coordinated cross-functional teams to gather business requirements and translate them into technical specifications
- Optimized Oracle 12g database performance, improving query response time by 35% through index optimization
- Created RPA solutions with Automation Anywhere and Python, reducing manual effort by 40%
- Developed complex workflows using Business Process Modeling Language (BPML) for enterprise clients
- Led technical design sessions and code reviews to ensure quality and adherence to best practices
- Received awards: Accenture Celebrates Excellence (2019, 2020), Tech Star 2021

KEY PROJECTS

Healthcare System Integration Hub

- Designed and implemented a centralized integration platform connecting disparate healthcare systems
- Implemented secure data exchange using healthcare standards (HL7, NCPDP)
- Coordinated testing and integration with third-party software vendors
- Documented system architecture and provided technical training to team members

Health Inequities in EPIC MyChart portal activation [[github](#)]

- Developed **React.js** dashboard with **D3.js** visualizations to analyze healthcare access disparities
- Built Node.js/Express backend with PostgreSQL database for analytics data storage and retrieval
- Implemented real-time filtering, demographic comparisons, and geospatial mapping features
- Dashboard insights increased MyChart adoption by **27%** among underserved populations
- **Presented the findings** at **UI Cancer Center**, demonstrating how **data visualization** can inform policy decisions

IBCCP Application for Breast Cancer Screening

- Architected Python/**React** application with **RESTful API microservices** and real-time RedCap integration
- Engineered advanced security features including **HIPAA-compliant JWT authentication**, encryption, and audit logging
- Built automated reporting system with customizable dashboards and scheduled PDF generation
- Deployed containerized application with **CI/CD pipeline**, resulting in **10-15 hours weekly** time savings

Cardiovascular Disease Visualization Platform [[github](#)]

- Built single-page application using **React**, **Redux**, and **D3.js** for interactive medical data visualization
- Optimized performance for handling **100,000+ patient records** with virtual scrolling and data chunking
- Designed intuitive interface allowing non-technical medical researchers to derive meaningful insights

Chicago Crime Index Application[[GitHub](#)]

- Developed React application with **Mapbox** integration for geographic crime visualization
- Created Python Flask API with PostgreSQL/**PostGIS** for spatial data analysis
- Built custom **D3.js** components for interactive crime statistics visualization and filtering

Email Assistant AI Workflow Automation

- Built a practical AI agent system that automatically categorizes, prioritizes, and drafts responses to customer support emails
- Implemented a simple but effective workflow automation pipeline using Python and Hugging Face transformers
- Achieved 92% accuracy in email classification and reduced response time by 65%